

Cooling Tower Water Efficiency & Legionella Control - Maintenance Record

Tanjong Quay Tower · Office · Singapore

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1. Maintenance log summary (2024)

Tanjong Quay Tower operates four induced-draft cooling towers serving a 2,400 RT chilled-water plant. Water treatment was manual: the contractor opened the blowdown valves twice a day and dosed chemicals by hand. Cycles of concentration averaged only 3.5 and make-up water use reached 62,000 m³ per year.

Date	Observation	Action taken
12 Feb 2024	High conductivity on CT-2 after a missed blowdown	Manual blowdown; dosing adjusted
07 May 2024	Legionella count above action level, CT-3	Tower isolated, disinfected, retested clear
19 Aug 2024	Scale on condenser tubes of chiller CH-2	Tubes brushed; approach temperature restored
03 Oct 2024	Legionella count above action level, CT-1	Disinfected; damaged drift eliminators found

Water treatment contractor on record: Mr Tan Wei Ming (mobile 9123 4567, weiming.tan@example.com).

2. Corrective actions

- Automated conductivity controller with a motorised blowdown valve on each tower; blowdown now happens continuously in small amounts instead of twice a day.
- Cycles of concentration raised from 3.5 to 6 with a scale and corrosion inhibitor programme and corrosion coupon monitoring.
- Side-stream filtration added to remove suspended solids.
- Automated biocide dosing controlled by oxidation-reduction potential (ORP).
- Legionella sampling increased from quarterly to monthly for each tower; damaged drift eliminators replaced.
- Conductivity, ORP and make-up water meters integrated into the BMS, with alarms to the control room.

3. Results (18 months)

KPI	2024 baseline	2025-2026 (annualised)
Make-up water (m³/yr)	62,000	49,000
Cycles of concentration	3.5	6.0
Legionella exceedances	2	0
Chemical cost	baseline	-9%

4. Lessons learned

- Automation removes the variability of manual blowdown; most exceedances followed missed or late manual actions.
- Blowdown valves must be fail-safe and must alarm when stuck open or stuck closed.
- Higher cycles need tighter corrosion monitoring; coupons are cheap insurance.
- Coordinate with chiller plant optimisation: a lower condenser-water temperature saves chiller energy but increases evaporation.